# COMP411/511: Project Report

**Deadline: 11.00 am - January, 20th**

**Ömer Faruk Doğançay, Mahmut Zahid Göksu , Ömer İslam, Berk Ömürlü**

## 1 Introduction

- **Problem Definition:** This project will benchmark four trending depth perception models –Depth Anything v2, Marigold, Depth Pro and DepthCrafter– to evaluate their effectiveness for a Mars rover application. Depth perception is crucial for autonomous navigation, as it enables the rover to safely navigate Mars's challenging terrain by accurately measuring the distances and identifying obstacles. In this project, Mars surface simulations will be utilized to create a realistic dataset, enabling to evaluate both the pretrained models and Marigold's fine-tuned version specifically for Martian environment.

- **Motivation:** This project is essential because accurate depth perception is a cornerstone of a safe and autonomous Mars exploration. The Martian environment presents unique navigation challenges, including irregular terrain, steep inclines, and dust interference. Benchmarking Depth Anything v2, Depth Pro, and DepthCrafter using their pretrained version and the Mars-specific fine-tuned adaptation of Marigold will yield insights that could inform the design of future autonomous systems for space exploration. Additionally, this project brings together cutting-edge deep learning in Computer Vision with real-world applications in planetary exploration. Tackling depth perception for a Mars rover application not only advances this technology but also contributes to the broader goal of interplanetary autonomy, enhancing the future Mars missions safety and efficiency.

- **Contributions:**
  - Simulate Mars terrain and create a dataset for training and testing.
  - Fine-tune a pre-trained model, Marigold, on this synthetic dataset.
  - Benchmark each model using metrics proposed in their respective research papers, but on custom proposed Mars-like surfaces.
  - Compare the pre-trained and fine-tuned models to assess performance improvements on Mars-like surfaces.

## 2 Related work

- **Starting Point:** This project builds on recent advancements in depth perception, particularly in unstructured and complex terrains similar to Mars. Baseline models like DepthAnything v2, Marigold, Depth Pro, and DepthCrafter have shown promising results in depth estimation across diverse environments on Earth. These models are pretrained on extensive datasets and employ various deep learning architectures that capture fine-grained details in visual depth cues. This baseline work is chosen because these models have robust depth perception capabilities, essential for a Mars rover navigating challenging terrain. Fine-tuning the Marigold model on a simulated Mars environment aims to assess and adapt its capability for an extraterrestrial environment, directly aligning with optimizing depth perception for a Mars rover application. In recent years, depth perception research has seen substantial progress, particularly with methods utilizing self-supervised

learning, monocular depth estimation, and domain adaptation and zero-shot learning. Challenges in monocular depth estimation are: relative depth estimation and metric depth. Relative depth estimation is the task of ordering each object depth-wise where metric depth estimation is predicting the absolute distance of each scene point from the camera. A middle ground to these two depth estimation methods have been proposed: addine-invariant depth which is equivalent to metric depth up to an unknown global scale and shift of the scene. Unlike metric depth, affine-invariant depth does not depend on the focal length, making it easier to predict in unfamiliar scenes where there are no objects that can serve as metric references. However, affine-invariant depth still maintains the relative distance ratios between objects. Metric depth can be derived from affine-invariant depth by anchoring it with a few known depth points or by estimating the missing scale and offset directly.

- **Related Work:** Recent challenges in depth estimation have been generalization of models. Training a depth estimator across diverse datasets has shown to generalize depth prediction with MegaDepth. Later MiDaS advanced this with scale and sift invariant loss with these diverse datasets. Later work has shown that using transformer based architectures help further improve the capabilities utilizing unlabeled datasets and training in a self-supervised setting [1]. Many models harness diffusion models to synthesize relative depth maps demonstrating excellent generalization but are ambiguous in scale and shift. Metric depth prediction has been improved through global depth value distribution and scene conditioning, considering camera intrinsics. For example, Cam-Convs integrates camera intrinsics, metric 3D scales images or depth maps to canonical space and remaps the estimated depth given the focal length and ZeroDepth learns camera-specific embeddings in a variation framework [1]. These models require the camera intrinsics to be known and accurate. There have been attempts to reason about these unknown camera intrinsics either by using a separate network or by predicting a camera intrinsic for conditioning its depth predictions in spherical space. Vision transformers have emerged as the ruling general architecture for perception tasks. Latest models for depth estimation utilized the basic ViT architecture. Depth Pro utilizes this architecture using a ViT encoder-decoder framework. On the other hand Marigold uses Latent Diffusion models to enhance their depth estimation and map generation [3]. Similar to Marigold, DepthCrafter has proposed a diffusion based model for depth estimation [2]. These models have been trained for the affine-invariant depth and metric depth. DepthAnything V2 on the other hand uses a teacher student framework utilizing ViT based models trained on purely synthetic data to improve generalization and for precise measurements and unlabeled real images to produce precise pseudo depth on large-scale unlabeled [4].

# 3 Method

- **Overview:** Using Unreal Engine 5, we created a Mars surface resembling, trying to replicate the topology and the textures, using available community assets. UE5 provides a visual scripting language that allows to map depth buffer, and the final graphic output to a texture. Then we started the simulation and moved the camera as we pleased. During runtime we save the screenshots of these textures as .exr files, which has the raw value of depth buffer in its R channel.

  After that we preprocessed each data point to fit the models' requirements. We had trouble working with Marigold as it required much more gpu ram. At the end we had to work with cropped images as images with 1920x1080 resolution did not fit 40 GB A100 gpu's. With the preprocessing we have paid careful attention to the hyperparameters and the preprocessing steps for pretraining of the model to get better results in training and inference. For other models we have used well known metrics such as f1, threshold s1 and absRel and benchmarked them on our custom data.

- **Model Details:**

  - **Marigold:** The article Repurposing Diffusion-Based Image Generators for Monocular Depth

Estimation, published in December 2023, was the first to introduce the Marigold model. Marigold differs from conventional depth estimation techniques. It incorporates diffusion-based models into the depth estimation domain, commonly employed for image generation [3]. The researchers had the following idea: Could image generation models be used for depth estimation if they have already learned high-quality images from different domains posted online [3]? Marigold thus makes use of Stable Diffusion's pre-trained capabilities [3]. This generative model needs to be fine-tuned to be used for depth estimation. The fine-tuning structure, as shown in Figure 1, processes the depth map, the actual image, and their corresponding latent space encodings using a variational autoencoder [3]. After adding noise to the depth map, the system concatenates the noisy map with the actual image [3]. The diffusion model then rebuilds the depth map after eliminating the noise [3]. This method modifies the latent diffusion model to create depth maps, which serves as the basis for stable diffusion [3]. The researchers employ synthetic data to enhance training performance [3]. They prefer synthetic data over datasets with real depth values because it ignores the physical limitations that frequently decrease the accuracy of real-world data [3]. The inference structure goes through a similar procedure after fine-tuning is finished. A high-resolution depth map is created by decoding the final image after noise has been added and removed [3].

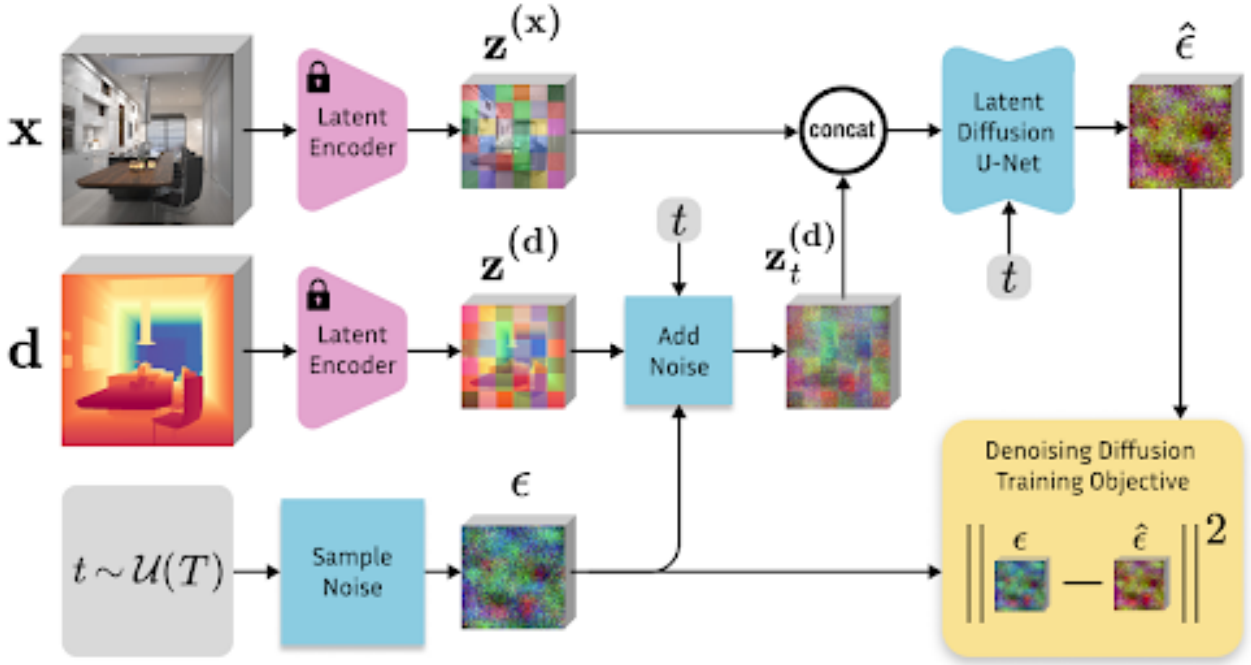



Figure 1: Overview of the Marigold fine-tuning protocol[3]

– **Depth Anything v2:** Researchers introduced Depth Anything v2, a monocular depth estimate model, in June 2024 with the tagline, "without pursuing fancy techniques." [4]. This model can handle complicated scenarios with transparent objects and reflections, generate extremely accurate depth maps with fine details, and generalize well to a variety of domains by combining the benefits of synthetic and real data. There are three primary steps in the model: First, only precise synthetic images are used to train a very powerful teacher model built on the DINOv2-Giant encoder [4]. This guarantees that the model will properly learn to predict fine-grained depth details. The pseudo depth labels for a large number of unlabeled real images are then generated using this teacher's model [4]. Lastly, the pseudo-labeled real data is used to train smaller student models of various scales [4]. This gives the students the

opportunity to learn robust depth estimation while taking use of the real data distribution and the teacher's predictions.

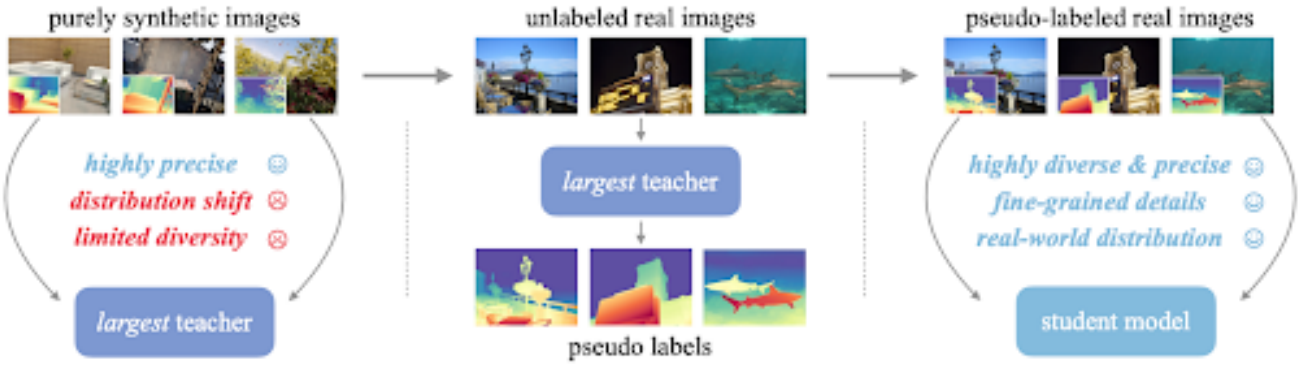



Figure 2: Overview of the Depth Anything V2[4]

– **Depth Pro:** Developed by Apple's AI research team, Depth Pro is a foundation model for zero-shot metric monocular depth estimation that generates high-resolution depth maps precisely and with fine details without requiring information like camera intrinsics. To create a single high-resolution depth map, Depth Pro processes image patches at various sizes using ViT encoders and fuses the predictions [3]. For scale-invariant learning, it uses a patch encoder; for global context, it uses an image encoder [3]. The input image is downsampled and separated into patches. To prevent seams, the patches overlap at finer scales and due to their independent processing, these patches bring both computational efficiency and parallelization [3]. As researchers also mentioned in their paper, because the difficulty of multi-head self-attention increases quadratically with input pixels, the patch-based architecture of the network is more efficient than scaling the ViT directly [3].

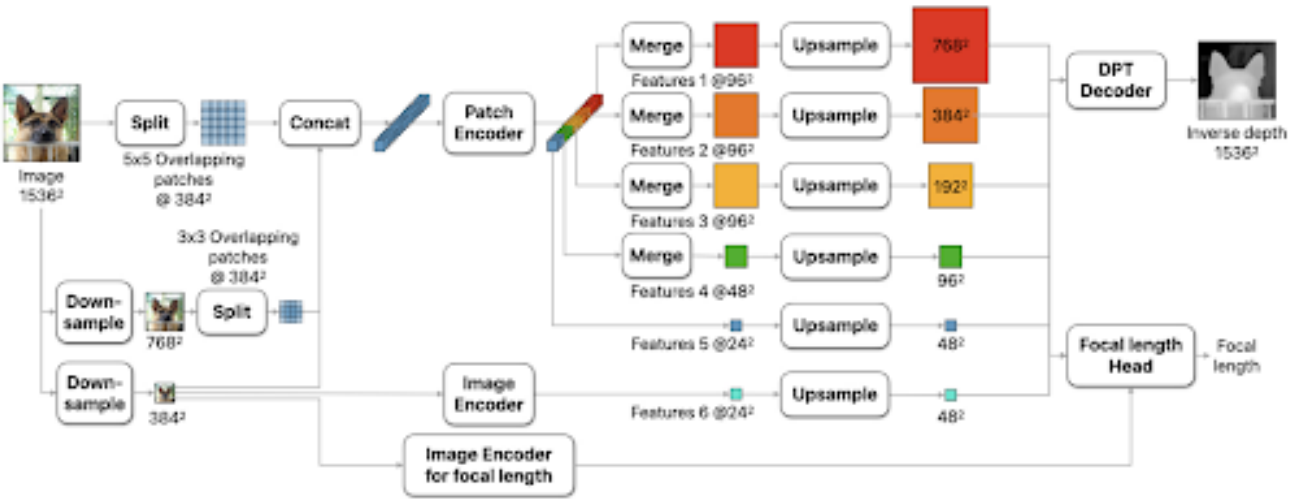



Figure 3: Overview of the Depth Pro's network architecture[1]

– **DepthCrafter:** Depth Crafter was introduced by Tencent AI Lab in September 2024 as a "novel video depth estimation approach that can generate temporally consistent long depth sequences" [2]. Basically, DepthCrafter simulates the distribution over the depth sequence based on the input video. The model was trained in three steps, using assembled realistic or synthetic datasets of different durations T to gradually learn the diffusion model's spatial

or temporal layers [2]. Given an open-world video, it uses initialized Gaussian noise to infer temporally consistent long depth sequences with fine-grained features over the duration of the video without the need for additional information like camera positions or optical flow [2].

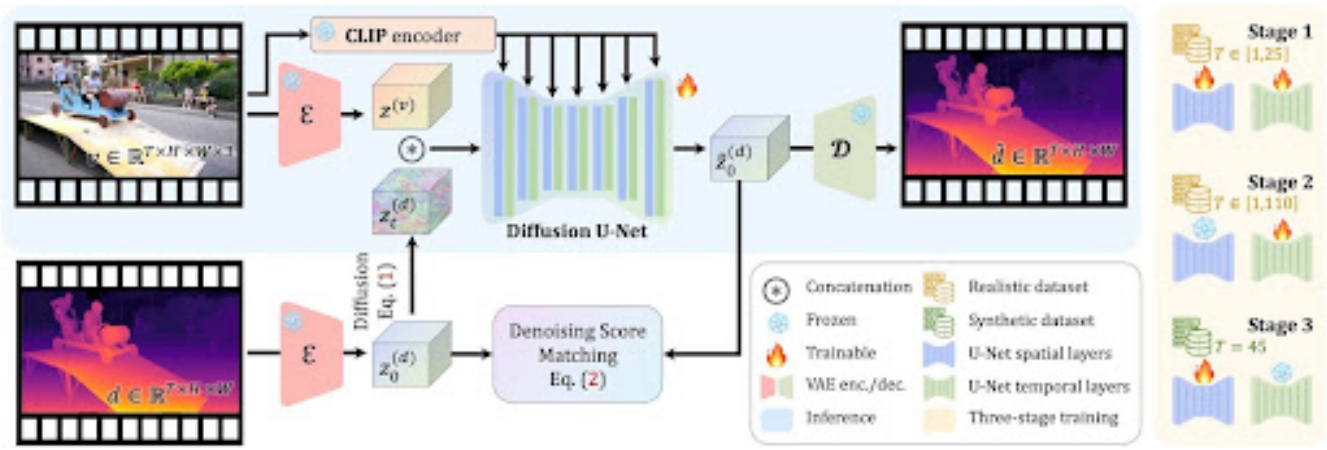



Figure 4: Overview of the Depth Crafter[2]

# 4 Results

- **Datasets and Metrics:** The metrics we have used are f1 which is essentially a mean of overestimation and the underestimation of the depths, as for the function itself is taken from the Depth Pro. Absolute relative difference is the second metric we have used to measure the average distance difference between our prediction with the ground truth. As for the last metric, we have used delta 1 which measures the percentage of predictions with a disparity from the ground truth higher than an epsilon, which is 1.25 in our case. Implementation of both delta 1 and absolute relative difference is taken from the Marigold's evaluation metrics.

- **Evaluation Protocol:** For the evaluation process we have run inferences on all models without training using the metrics mentioned above. Using all the data set we have measured the F1, Absolute Relative Distance and the Delta 1 values. The evaluations are conducted on T4 Gpu for Depth Crafter, Depth Anything and Marigold, whereas A100 is used for the Depth Pro due to the high resolution requirement of the model with size 1536 x 1536. As for the model specific evaluations, the dataset is converted to video format for the Depth Crafter as it requires videos as input. After the inference step the output video is processed frame by frame in order to apply the evaluation metrics. Furthermore, for the evaluation our ground truth is converted to size 576 x 1024 from 1920 x 1080, which is the highest resolution available in Depth Crafter. For all the remaining models, the same protocol of applying the evaluation metric after each model output is used.

- **Results:**

| Model | Mean | Median | Standard Deviation |
|---|---|---|---|
| Depth Anything V2 | 0.0367 | 0.0247 | 0.0353 |
| DepthCrafter | 0.0045 | 0.0001 | 0.0231 |
| Depth Pro | 0.0907 | 0.0868 | 0.0623 |
| Marigold | 0.0236 | 0.0140 | 0.0241 |
| Marigold (Fine-tuned) | 0.0289 | 0.0272 | 0.0256 |

Table 1: F1 Statistics

| Model | Mean | Median | Standard Deviation |
|---|---|---|---|
| Depth Anything V2 | 4501.41 | 4696.83 | 1805.88 |
| DepthCrafter | 1591.62 | 1382.38 | 849.70 |
| Depth Pro | 9.03 | 6.65 | 14.78 |
| Marigold | 52151.66 | 32255.34 | 105680.92 |
| Marigold (Fine-tuned) | 41347.63 | 39512.95 | 7934.22 |

Table 2: Absolute Relative Error Statistics

| Model | Mean | Median | Standard Deviation |
|---|---|---|---|
| Depth Anything V2 | 0.3760 | 0.3727 | 0.0638 |
| DepthCrafter | 0.0000 | 0.0000 | 0.0000 |
| Depth Pro | 0.0077 | 0.0000 | 0.0313 |
| Marigold | 0.2760 | 0.2695 | 0.1434 |
| Marigold (Fine-tuned) | 0.3445 | 0.2734 | 0.1223 |

Table 3: Delta 1 Statistics

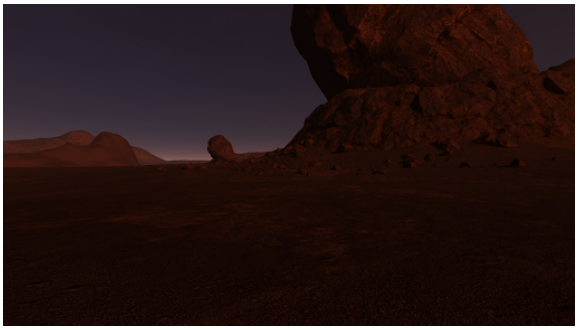
| Method | Booster | ETH3D | Middlebury | NuScenes | Sintel | Sun-RGBD | Avg. Rank↓ |
|---|---|---|---|---|---|---|---|
| DepthAnything (Yang et al., 2024a) | 52.3 | 9.3 | 39.3 | 35.4 | 6.9 | 85.0 | 5.7 |
| DepthAnything v2 (Yang et al., 2024b) | 59.5 | 36.3 | 37.2 | 17.7 | 5.9 | 72.4 | 5.8 |
| Metric3D (Yin et al., 2023) | 4.7 | 34.2 | 13.6 | 64.4 | 17.3 | 16.9 | 5.8 |
| Metric3D v2 (Hu et al., 2024) | 39.4 | 87.7 | 29.9 | 82.6 | 38.3 | 75.6 | 3.7 |
| PatchFusion (Li et al., 2024a) | 22.6 | 51.8 | 49.9 | 20.4 | 14.0 | 53.6 | 5.2 |
| UniDepth (Piccinelli et al., 2024) | 27.6 | 25.3 | 31.9 | **83.6** | 16.5 | **95.8** | 4.2 |
| ZeroDepth (Guizilini et al., 2023) | OOM | OOM | 46.5 | 64.3 | 12.9 | OOM | 4.6 |
| ZoeDepth (Bhat et al., 2023) | 21.6 | 34.2 | 53.8 | 28.1 | 7.8 | 85.7 | 5.3 |
| Depth Pro (Ours) | 46.6 | 41.5 | **60.5** | 49.1 | **40.0** | 89.0 | **2.5** |

Figure 5: Zero Shot Metric Depth Accuracies and Delta 1 Statistics from Depth Pro: Sharp Monocular Metric Depth in Less Than a Second[1]

| | Method | Sintel F1↑ | Spring F1↑ | iBims F1↑ | AM R↑ | P3M R↑ | DIS R↑ |
|---|---|---|---|---|---|---|---|
| Absolute | DPT (Ranftl et al., 2021) | 0.181 | 0.029 | 0.113 | 0.055 | 0.075 | 0.018 |
| | Metric3D (Yin et al., 2023) | 0.037 | 0.000 | 0.055 | 0.003 | 0.003 | 0.001 |
| | Metric3D v2 (Hu et al., 2024) | 0.321 | 0.024 | 0.096 | 0.024 | 0.013 | 0.006 |
| | ZoeDepth (Bhat et al., 2023) | 0.027 | 0.001 | 0.035 | 0.008 | 0.004 | 0.002 |
| | PatchFusion (Li et al., 2024a) | 0.312 | 0.032 | 0.134 | 0.061 | 0.109 | 0.068 |
| | UniDepth (Piccinelli et al., 2024) | 0.316 | 0.000 | 0.039 | 0.001 | 0.003 | 0.000 |
| Rel. | DepthAnything (Yang et al., 2024a) | 0.261 | 0.045 | 0.127 | 0.058 | 0.094 | 0.023 |
| | DepthAnything v2 (Yang et al., 2024b) | 0.228 | 0.056 | 0.111 | 0.107 | 0.131 | 0.056 |
| | Marigold (Ke et al., 2024) | 0.068 | 0.032 | 0.149 | 0.064 | 0.101 | 0.049 |
| | Depth Pro (Ours) | **0.409** | **0.079** | **0.176** | **0.173** | **0.168** | **0.077** |

Figure 6: Zero Shot Boundary Accuracies and F1 Statistics from Depth Pro: Sharp Monocular Metric Depth in Less Than a Second[1]

- **Ablations and Experiment Conclusions:** Initially, we aimed to fine-tune all the models and started with training Depth Pro. However, upon training with a couple of images we have realized that the architecture used requires too much GPU power and is not feasible to be trained. We have also tried data augmentation with adding brightness to the image in the inference step, however the results did not show any significant improvement and was very close to the original image.



(a) Original Image

(b) Predicted Depth Map Before Fine-Tuning

(c) Predicted Depth Map After Fine-Tuning

Figure 7: Comparison of Original Image and Predicted Depth Maps Before and After Fine-Tuning

# 5 Conclusion

At first we considered using the "Mars 2020 Terrain Relative Navigation HiRISE DTM Mosaic" data set published by NASA. This data was originally gathered to provide hazard avoidance for the Mars 2020 rover that would explore Jezero crater. However it operates with a 0.3-meter/pixel resolution for its raw imagery from orbit, but creating DTMs involves stereo imaging and interpolation, which slightly reduces the effective resolution to 1 meter/pixel for terrain models. This wasn't enough for our use cases as our purpose is to simulate a mars rover's navigation, not the landing phase. Our use case requires much higher resolution on the horizontal axis.. In the future we want to reconstruct this DTM in a 3d representation and then render it using Martian lighting settings. We would need to apply textures with a reasonable variance and fidelity to be as close as possible to reality. If we can successfully complete these steps it would be the most comprehensive dataset for computer vision in the context of Martian exploration. Since these steps require comprehensive understanding of HiRISE and a large amount of intensive manual labour, we concluded that this methodology would be a data engineering project rather than a computer vision benchmark. These constraints made us consider purely synthetic data. Thus we decided to generate our data set using Unreal Engine 5 and available community assets.

# 6 Acknowledgments


- **Ömer Faruk Doğançay:** Researched available data sets, evaluated feasibility of each one. Created the setup and executed the simulation to generate the synthetic data. Handled preprocessing and post processing pipelines of the data and evaluations.
- **Mahmut Zahid Göksu:** Inspected pipelines of the models, instructed the team to correctly set the foundations for evaluation and fine tuning processes. Implemented fine tuning of Marigold model and completed the evaluation processes for both the fine tuned and raw model.
- **Ömer İslam:** Compiled the evaluation methods to be used in the inference, as well as conducted all the evaluation pipelines for the Depth Pro, Depth Crafter and Depth Anything.
- **Berk Ömürlü:** Researched the models we used in the project in detail and learnt their working principles and details, assisted the team with said research whenever required, reported the models' inner workings to the team and for the final report.

# 7 References

# References


[1] Aleksei Bochkovskii, Amaël Delaunoy, Hugo Germain, Marcel Santos, Yichao Zhou, Stephan R. Richter, and Vladlen Koltun. Depth pro: Sharp monocular metric depth in less than a second. *arXiv*, 2024.

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[3] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing diffusion-based image generators for monocular depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

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